

PROJECT GUIDE

AI Invoice Processing Agent for Accounting Teams

A simple, automated system for handling invoices from email to payment

Built with: n8n, OpenRouter (AI), Gmail, Airtable, Slack, Google Calendar

Prepared by: AI Automation Engineer

For: Employers, AI agencies, recruiters, and clients

1. What This Project Does

This is an automated system that handles invoices from start to finish — no manual data entry needed. It watches an email inbox, finds invoice emails, reads the attached PDF, pulls out the important details using AI, saves everything to a database, alerts the finance team, and reminds them before payments are due.

Quick Summary

- Reads invoices automatically — no manual typing
- Stops duplicate payments before they happen
- Reminds the team 3 days before a payment is due
- Tracks every invoice as Pending, Paid, or Overdue

2. The Problem It Solves

Most accounting teams handle invoices by hand: opening emails, reading PDFs, typing numbers into spreadsheets, and trying to remember due dates. This is slow and easy to get wrong.

- **Slow:** Every invoice takes several minutes of manual work.
- **Error-prone:** Typing mistakes lead to wrong amounts or missed details.
- **Missed deadlines:** Due dates get lost in inboxes, leading to late fees.
- **Duplicate payments:** The same invoice can get paid twice by accident.
- **No visibility:** Managers can't easily see what's owed and when.

3. Why It's Worth Automating

Invoice handling is repetitive and rule-based — exactly the kind of task AI and automation handle well. Automating it saves time, reduces mistakes, and lets accounting staff focus on more valuable work like reviewing budgets instead of retyping numbers.

- Works 24/7, even outside office hours
- Same effort whether there are 10 invoices or 1,000
- Creates a clean, searchable record of every invoice

4. How It Works (Step by Step)

New invoice email arrives

|

v

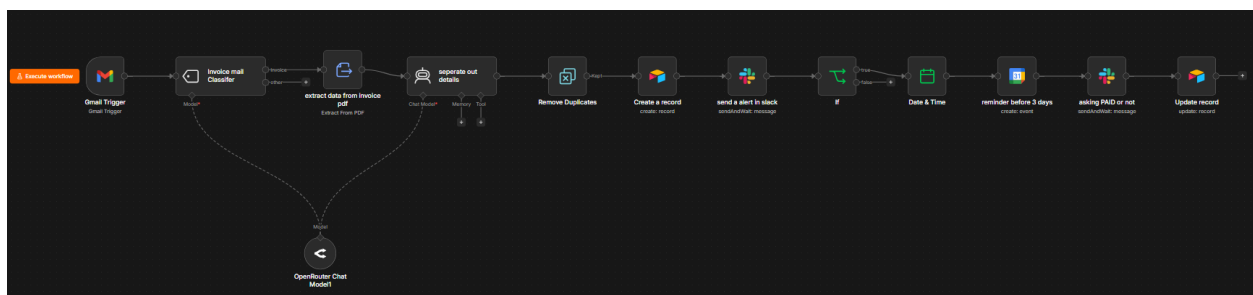
1. Check if it's really an invoice (AI)

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v

2. Pull out the PDF and read it (AI)
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 - v
3. Check it isn't a duplicate
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4. Save the invoice to Airtable
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 - v
5. Send a Slack alert to finance
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 - v
6. 3 days before due date: send a reminder
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 - v
7. Mark it Paid, Pending, or Overdue

This is the real workflow, built in n8n:



5. What Data Is Collected

The AI reads each invoice and pulls out these details automatically:

- Vendor name
- Invoice number
- Invoice date and due date
- Total amount and tax amount
- GST number
- Payment terms

Example of what the AI returns:

```
{
  "vendor_name": "ABC Solutions",
  "invoice_number": "INV-2026-101",
  "invoice_date": "2026-06-15",
  "due_date": "2026-07-15",
  "total_amount": 29500,
```

```
"tax_amount": 4500,  
"payment_terms": "Net 30"  
}
```

6. The Tools Used

Tool	What It Does
n8n	Connects every step and runs the automation
OpenRouter (AI)	Reads emails and PDFs, pulls out the invoice details
Gmail	Where invoices arrive
Airtable	Stores every invoice record
Slack	Alerts the finance team and asks for payment updates
Google Calendar	Schedules due-date reminders

7. Keeping Data Accurate

- **No duplicate payments:** Before saving, the system checks if that invoice number and vendor already exist. If it's already in the system, it's skipped.
- **Only real invoices get processed:** An AI check filters out newsletters and other emails first, so only actual invoices move forward.
- **Clean, structured data:** The AI is required to return information in a fixed format, so the rest of the system always gets data it can use.

8. Business Impact

	Manual Process	With This System
Time per invoice	5–10 minutes	A few seconds, no effort
Duplicate payments	Can happen	Prevented automatically
Due date tracking	Easy to forget	Automatic reminders
Team visibility	Have to check manually	Instant Slack alerts

For a team handling 200 invoices a month, this can save 15–30 hours of manual work every month — time the team can spend on more useful tasks.

9. Possible Improvements

- Add OCR support for scanned (image-based) invoices
- Connect directly to accounting software like QuickBooks or Xero
- Flag low-confidence AI extractions for a quick human check

- Build a simple dashboard showing total payables and overdue invoices

10. Skills Shown in This Project

- Building multi-step automations in n8n
- Using AI to read and extract data from documents
- Connecting multiple tools together (Gmail, Airtable, Slack, Calendar)
- Designing systems that prevent costly mistakes (like duplicate payments)
- Turning a real business problem into a working, useful solution

11. Conclusion

This project shows how AI and automation can take over a repetitive, error-prone task and turn it into a fast, reliable, self-running process. It combines AI (for reading documents) with simple, clear logic (for decisions and tracking) — giving a finance team a tool they can actually trust and use every day.